Guilherme Trindade

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WORK EXPERIENCE

Mercor August 2026 – Present

AI Safety Evaluation Contractor — Biology Expert (PhD) & Generalist Expert *Remote*

- Design adversarial and expert-level prompts to probe frontier LLM capabilities and elicit failure modes in specialized life-science domains and general-knowledge tasks.
- Annotate and grade model outputs against structured rubrics for factual accuracy, reasoning quality, helpfulness, and appropriate handling of dual-use content.
- Classify multi-turn conversations for policy-relevant risk signals, applying consistent judgment across ambiguous and borderline cases.
- Certified in adversarial robustness: jailbreak versus prompt injection, multi-turn escalation patterns, and red team evaluation design.

NYU Langone School of Medicine – Laboratory of Translational Obesity Research (LTOR) March 2024 – Sept 2025

Graduate Research, Alemán Lab *New York City, NY*

- Led a multi-omics study of macrophage inflammation in obese adipose tissue, integrating human and mouse bulk/scRNA-seq with LC-MS lipidomics in Python and R.
- Built a reproducible R pipeline on NYU HPC (FastQC → fastp → STAR → featureCounts → DESeq2 → mixOmics/MetaboAnalyst) with automated QC, batch correction, normalization, and RMarkdown/HTML reporting (on GitHub).
- Identified five monoglyceride hotspots and a TG (46:9) cluster; linked PPAR signaling via GSEA/KEGG and built gene-lipid networks in Cytoscape. Poster at ObesityWeek, San Antonio (2024).
- Integrated Claude AI into workflows to prototype pipelines, debug R/Python, and build interactive data visualization tools; coordinated postdocs, statisticians, and LC-MS engineers and mentored juniors.

Additional: NYU Langone Neurology (2024): Python clustering/ML on Alzheimer's disease data, IHC.

PROJECTS

OmicsHonesty: Open AI Honesty Eval for Bioinformatics (in progress) Solo Builder

- Building an open-source eval in UK AISI Inspect testing whether LLMs honestly report null results, impossible requests and silent pipeline failures instead of inventing findings under user pressure.
- Designed 5 starter tasks with a model-graded honesty rubric; next: known-answer tasks from public RNA-seq data and agentic tasks.

BioAgent: Agentic Bioinformatics Workbench Solo Builder

- Built a full-stack workbench, adopted by members of my lab, where a researcher describes an analysis in plain English and a Claude-powered agent plans and executes real bioinformatics tools: DESeq2, PCA, KEGG enrichment, PubMed search.
- Engineered a Python/FastAPI backend exposing six validated tools via Claude tool use, with a streaming React/TypeScript UI, per-run reproducibility manifests, and separation of computed results from model interpretation.

Seatd — AI Front Desk SaaS for Salons, Barbershops & Nail Studios Founder & Solo Builder

- Founded and built a SaaS automating the front desk for beauty & grooming businesses via AI-powered SMS: bookings, reminders, missed-call recovery, review requests, and SMS marketing.
- Engineered 8 production n8n workflows integrating Claude Sonnet, Twilio SMS, and Google Sheets; shipped a go-to-market stack: Vercel landing page, onboarding, Stripe billing, per-client number provisioning.

Lab Rotation, SCI:FI Lab (PI: Dr. Frank Krueger), George Mason University Fall 2026

- Built a reproducible Python pipeline that extracts and validates Edelman Trust Barometer data into a 33-country panel (2016 to 2026).
- Found the high- vs low-income trust gap widened from 8.5 to 11.9 points across 23 countries; factor analysis showed one shared factor explains 79% of variance in trust across business, government, media and NGOs.
- Set a forecasting baseline (year-over-year r = 0.95; a no-change model lands within 3 points), identified an untested gap in out-of-sample trust prediction, and drafted the request for respondent-level data.

SKILLS & ADDITIONAL INFORMATION

AI & Machine Learning: Agentic AI & LLM tooling (Claude API, tool use, agent loops, pipeline dev); LLM evaluation and red teaming (adversarial prompting, rubric-based grading); model building/tuning (k-NN, regularized regression, ensembles), PCA, factor analysis, cross-validation, forecasting

Programming: Python (NumPy, pandas, scikit-learn, FastAPI, Flask/Django), R (caret, tidyverse, Bioconductor), Bash/Linux CLI, SQL, TypeScript/React, Git/GitHub

Bioinformatics & Multi-omics: Bulk/scRNA-seq (DESeq2, STAR, featureCounts), lipidomics (LipidSearch, MetaboAnalyst, mixOmics), GSEA/KEGG, Cytoscape, NCBI/Ensembl APIs, HPC/Slurm

Languages: English (native), Portuguese (fluent), Spanish (conversational)

Community Service: The Bowery Mission, New York City (volunteer, twice monthly)

EDUCATION

George Mason University January 2026 – Present

PhD Student in Bioinformatics and Computational Biology | GPA: 4.0 *Fairfax, VA/Hybrid*

Relevant Coursework: AI and Deep Learning in Bioinformatics, Biological Data Analysis

New York University September 2023 – May 2025

M.S. in Quantitative and Computational Biology, Graduate School of Arts and Sciences *New York City, NY*

University of Delaware September 2019 – December 2022

B.S. in Biological Sciences, College of Arts and Sciences *Newark, DE*